UDMS — Complete System Specification

Field	Value
Product	UDMS — University Document Management System
Document type	Complete System Specification (SRS + Architecture + Features + LLD)
Date	August 2026
Classification	Proprietary

Document contents

1. System Requirements Specification (SRS)
2. Architecture Design
3. Features & Requirements
4. Low-Level Design (LLD) Workflows

Part 1 — System Requirements Specification (SRS)

UDMS — University Document Management System

Field	Value
Document ID	SRS-UDMS-001
Product	UDMS (University Document Management System)
Repository	`CDTS_Zamboanga`
Version	1.2
Date	August 2026
Classification	Proprietary

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for **UDMS** (**University Document Management System**): an enterprise multi-tenant platform for document registration, workflow automation, approvals, records desk operations, SLA tracking, audit trails, and public status tracking.

1.2 Scope

In scope

- Multi-tenant document lifecycle with unique tracking numbers
- Visual workflow designer and multi-step approvals (sequential, parallel, conditional)
- Records desk: request/release, digitization/OCR QA, receiving, archives, disposal, legal hold, retention
- SLA monitoring, notifications, search, AI help desk and document intelligence
- Subscription billing, tenant branding, platform super-admin
- Local auth, OAuth, MFA, and University Platform Access SSO (`AUTH_MODE=dual`)
- Public unauthenticated tracking for Public-visibility documents

Out of scope

- Full UHRIS, Library (ULMIS), or Inventory (UIMS) business logic (sibling platforms)
- Guaranteed 500-user / <3s search SLA until proven in staging load tests
- Physical RFID / biometric hardware

- veraPDF-certified PDF/A archival validation

1.3 Definitions

Term	Definition
UDMS	University Document Management System
Tracking number	Unique ID `{PREFIX}-YYYY-NNNNNNN` (default prefix `DTS`)
Tenant	Isolated organization on the shared SaaS database
SLA	Service-level agreement hours on workflow steps

1.4 References

- `CDTS_Zamboanga/README.md`
- `docs/DocsTrack-Complete-Guide.md` (legacy filename; product name is UDMS)
- `docs/engineering/ARCHITECTURE.md`
- `docs/API-Specs-Manual.md`
- Sibling: University Platform Access SSO topology

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2. Overall Description

2.1 Product Perspective

Monorepo with separately deployable processes:

Component	Stack	Role
API	AdonisJS 7, TypeScript, PostgreSQL, Redis	REST `/api/v1`
Web	React 19, Vite, Tailwind, ShadCN	Marketing, tenant app, admin, public track
Workers	BullMQ	Notifications, SLA checks, document intelligence
Storage	S3-compatible	Attachments
Search	Meilisearch (optional) + Postgres FTS	Full-text / OCR search

2.2 User Classes

Role	Description
Tenant Owner	Full org control including billing and custom roles
Admin	Day-to-day user/workflow/org administration

Manager / Approver	Supervisory review and approval queue
Staff / Member	Create, edit, submit documents
Viewer	Read-only access to non-draft documents
Records Officer	Digitization, receiving, archives, retention, legal hold
Super Admin	Cross-tenant platform administration
Public visitor	Track public documents without login

2.3 Constraints

- Shared-database multi-tenancy with `tenant_id` isolation
- Workers must run separately from the API process
- University Platform Access remains a separate deployment; UDMS verifies platform JWTs via JWKS
- Default max attachment size 25 MB; plan quotas enforced
- Submit allowed from Draft only (returned documents follow admin resubmit path)

2.4 Assumptions

- SMTP configured for invitations and email notifications
- Object storage credentials available in production
- Enterprise deployments wire University Platform Access SSO before retiring local-only login
- Approvers monitor the approval queue; SLA hours reflect operational norms

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3. Functional Requirements

3.1 Documents (FR-DOC)

ID	Requirement	Priority
FR-DOC-001	Assign unique tracking numbers `{PREFIX}-YYYY-NNNNNNN` per tenant sequence	Must
FR-DOC-002	Support lifecycle: draft → pending_approval → approved/rejected/returned → completed → archived	Must
FR-DOC-003	Restrict draft visibility to creator until submit	Must
FR-DOC-004	Version content/files (v0.x drafts, v1+ post-approval)	Must
FR-DOC-005	Support Public/Private visibility; Public enables `/track/{trackingNumber}`	Must

FR-DOC-006	Support Confidential/Restricted levels with watermarked downloads where applicable	Should
FR-DOC-007	Allow multipart and presigned uploads; enforce size and storage quotas	Must
FR-DOC-008	Support NAP-aligned retention schedules and legal hold blocking disposal	Must
FR-DOC-009	Queue async OCR/intelligence jobs on upload	Should
FR-DOC-010	Optional e-sign integration when enabled	Could

3.2 Workflows & Approvals (FR-WF / FR-APR)

ID	Requirement	Priority
FR-WF-001	Represent workflows as directed graphs with sequential, parallel, and conditional steps	Must
FR-WF-002	Allow create draft, publish, and archive workflows	Must
FR-WF-003	Auto-provision a default published workflow on submit if none exists	Must
FR-WF-004	Assign steps by role, named user, or department head	Must
FR-APR-001	Provide Approvals queue with Approve, Return, Reject (comment required), Forward	Must
FR-APR-002	Record routing timeline with actor and timestamp	Must

3.3 SLA (FR-SLA)

ID	Requirement	Priority
FR-SLA-001	Support optional SLA hours per workflow step	Must
FR-SLA-002	Periodically detect breaches (worker) and notify supervisors	Must
FR-SLA-003	Surface overdue/at-risk metrics on dashboard and reports	Should

3.4 Records Desk (FR-REC)

ID	Requirement	Priority
FR-REC-001	Record access requests (view/download/physical) with release desk workflow	Must
FR-REC-002	Digitization batches with OCR QA checklist before link	Must
FR-REC-003	Receiving log with channels and printable acknowledgment	Must
FR-REC-004	Archival transfer packages and disposal with certificates	Must

FR-REC-005	Legal hold blocks disposal	Must

3.5 Search, Notifications, Audit, AI

ID	Requirement	Priority
FR-SRH-001	Command palette and full search across documents, files, OCR text, users, workflows	Must
FR-NOT-001	Notify on assignment, approval outcomes, and SLA breaches via email/in-app/push	Must
FR-AUD-001	Append-only audit logs for security and compliance actions	Must
FR-AI-001	Permission-aware in-app Help Desk copilot; confirm before destructive actions	Should

3.6 Multi-Tenant, Billing, Auth

ID	Requirement	Priority
FR-MT-001	Isolate all tenant-owned data by `tenant_id`	Must
FR-MT-002	Self-service registration creates tenant + owner and onboarding	Should
FR-BIL-001	Subscription plans with user/document/storage limits and Stripe/PayMongo checkout	Should
FR-ID-001	Local JWT auth, optional OAuth (Google/Microsoft), optional TOTP MFA	Must
FR-ID-002	Support University Platform Access SSO via JWKS (`AUTH_MODE=dual`)	Must
FR-PUB-001	Rate-limited public tracking for Public documents	Must

3.7 Administration & Reports

ID	Requirement	Priority
FR-ADM-001	Manage users, roles, departments, document types, branding	Must
FR-ADM-002	Dashboard and reports with metrics and CSV export	Must
FR-ADM-003	Super-admin portal for tenants, plans, and platform analytics	Must

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4. External Interface Requirements

4.1 User Interfaces

Surface	Audience
Marketing / login / register	Public
`/app/*` tenant application	Authenticated users

`/admin/*` platform portal	Super Admin
`/track/*` public tracking	Unauthenticated

4.2 Software Interfaces

Interface	Detail
REST API	HTTPS JSON `/api/v1/*` ; Bearer JWT; optional `X-Tenant-ID`
OpenAPI / Swagger	`/api/v1/docs`
University Platform Access SSO	JWKS verification; launcher handoff to `/sso/callback#access_token=…`
Billing webhooks	Stripe / PayMongo / stub
SMTP / FCM / Socket.IO	Email, push, realtime notifications
S3 / Meilisearch / OCR-AI providers	Storage, search, intelligence

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5. Non-Functional Requirements

ID	Category	Requirement
NFR-SEC-001	Security	HTTPS in production; VineJS validation; Shield XSS/CSRF; Redis rate limits
NFR-SEC-002	Security	RBAC on sensitive routes; secrets in env/Secrets Manager; failed-login lockout
NFR-SEC-003	Security	Presigned file URLs only; optional virus scan flags
NFR-PERF-001	Performance	Target authenticated search p95 < 3s under ~500 concurrent users (aspirational until proven)
NFR-AVL-001	Availability	Liveness/readiness health endpoints; API and worker separation
NFR-AVL-002	Availability	Target RPO ≤15 min; documented RTO for rollback/restore/DR
NFR-CMP-001	Compliance	Immutable audit trail; retention schedules; disposal certificates; legal hold
NFR-UX-001	Usability	Modern browsers; responsive layout; permission-based navigation
NFR-OBS-001	Observability	Optional Prometheus `/metrics` , Sentry, structured logs

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6. Data Requirements

- PostgreSQL with UUID primary keys, soft deletes, ~40 tables
- Core domains: tenant config, auth/RBAC, documents/files, workflows/approvals, SLA, billing, notifications, audit, records desk
- Redis for cache, queues, and rate limiting

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7. Verification Criteria (Summary)

Scenario	Expected result
Create & submit document	Tracking number assigned; workflow starts; status pending_approval
Approve / return / reject	Timeline updated; notifications sent
Public track	Public document status visible without login
Legal hold	Disposal blocked
University Platform Access SSO	Platform JWT accepted in dual mode
Plan limit	User creation blocked at max users

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8. Document Control

Version	Date	Description
1.2	August 2026	SRS baseline; product branded as UDMS

Related design docs

- **UDMS-Architecture.md** — Architecture design (context, containers, layers, deployment)
- **UDMS-Features-Requirements.md** — Feature catalog + requirements matrix
- **UDMS-LLD.md** — Low-level design workflows (auth, document pipeline, SLA, records desk, async workers)

End of SRS — UDMS (University Document Management System)

Part 2 — Architecture Design

UDMS — University Document Management System

Field	Value
Document ID	AD-UDMS-001
Product	UDMS
Companions	UDMS-SRS.md · UDMS-Features-Requirements.md · UDMS-LLD.md
Date	August 2026

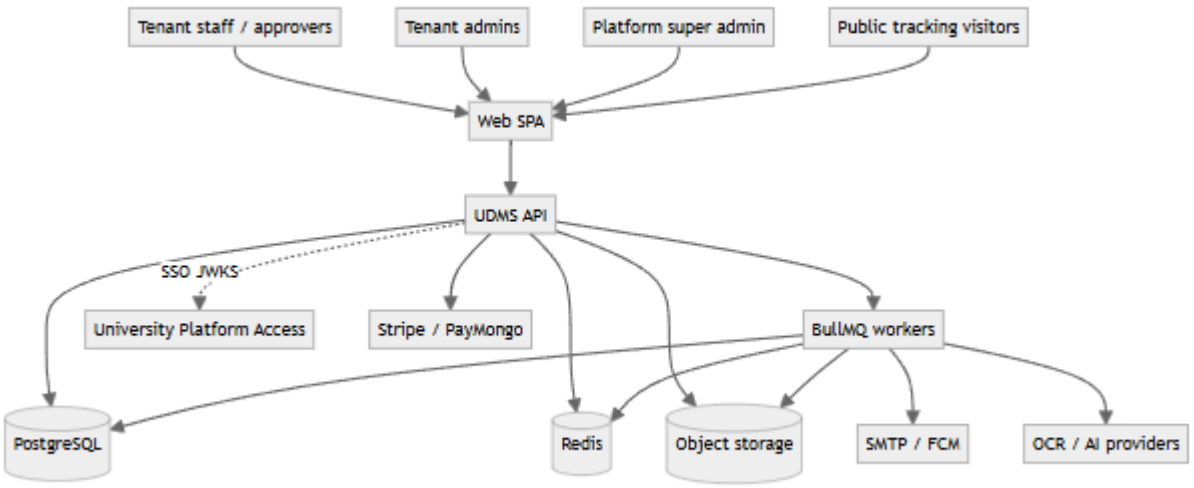
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1. Purpose

This Architecture Design (AD) describes the **system context, containers, layers, data stores, and deployment topology** for UDMS. Runtime workflows are in the LLD; requirements are in the SRS / Features catalogs.

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2. System context



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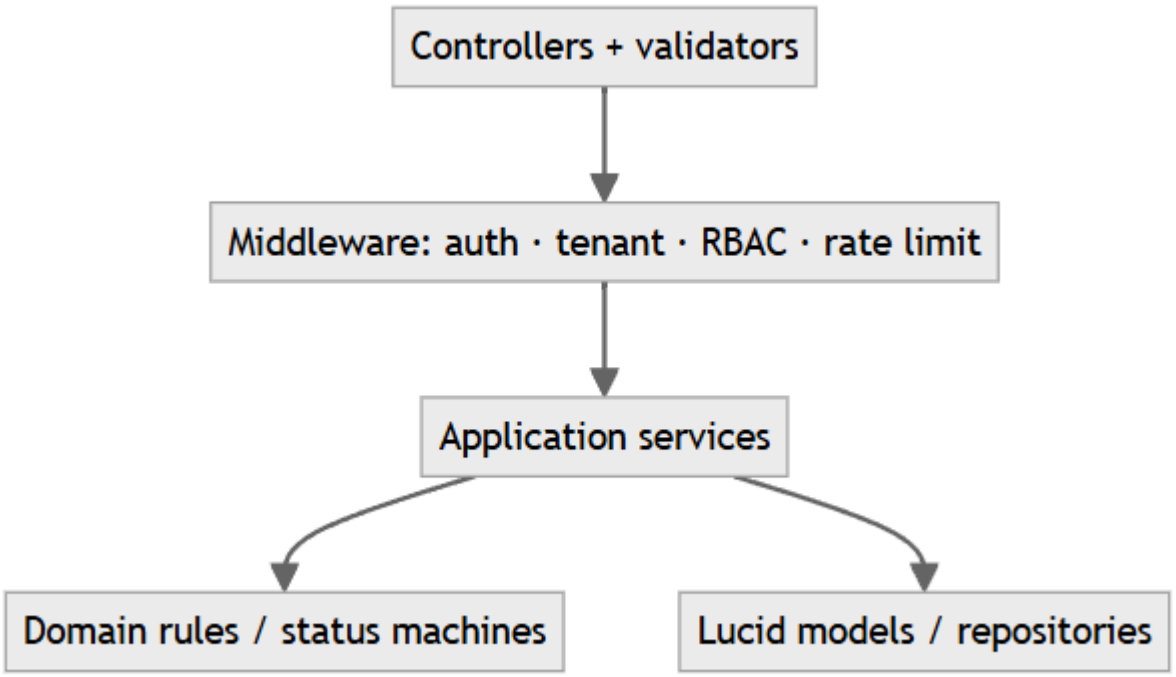
3. Container view

Container	Technology	Responsibility
Web	React 19, Vite, Tailwind, ShadCN	Marketing, `/app`, `/admin`, `/track`
API	AdonisJS 7, Lucid	Auth, RBAC, documents, workflows, records desk
Worker	BullMQ	Notifications, SLA checks, document intelligence
PostgreSQL	UUID PKs, soft deletes, `tenant_id`	System of record (~40 tables)
Redis	Cache + queues + rate limits	Async and performance
Object storage	S3 / R2 / MinIO	Attachments via presigned URLs
Search	Meilisearch (optional) + PG FTS	Indexed document/OCR search

Local ports (typical): API separate from UHRIS/ULMIS (e.g. 3335); Web Vite; Redis/Postgres via Compose.

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4. Backend layered architecture



Layer	Owns
Presentation	HTTP, OpenAPI, transformers
Application	DocumentService, WorkflowEngine, billing, records desk
Domain	Lifecycle enums, SLA rules, permission codes

Infrastructure	PG, Redis, S3, Meilisearch, mail, OCR adapters

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5. Multi-tenancy & security

- Shared-database isolation with `tenant_id` on tenant-owned rows
- Tenant resolved from user membership, verified domain, or `X-Tenant-ID`
- Bearer JWT (local) and/or University Platform Access RS256 JWT (`AUTH_MODE=dual`)
- RBAC via role_permissions; owners bypass within org
- Files only via presigned URLs; secrets in env / Secrets Manager

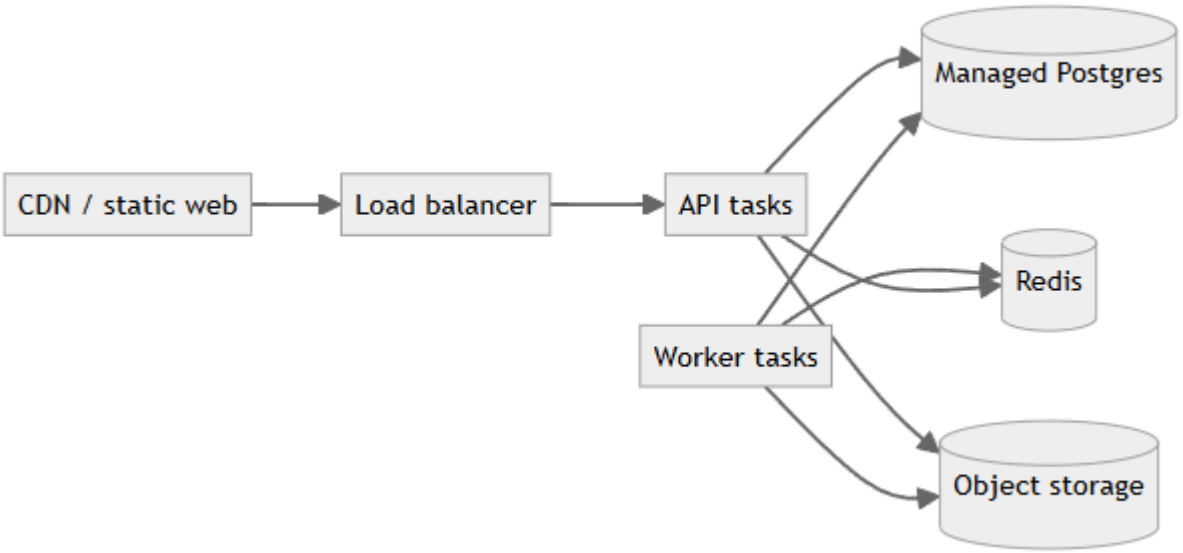
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6. Key domain modules

Module	Primary tables / concerns
Auth / RBAC	users, roles, permissions, sessions, MFA
Documents	documents, files, versions, tracking_sequences
Workflows	workflows, workflow_steps, document_workflow_instances, approvals
Records desk	record_requests, digitization, receiving, archives, retention
SLA	deadlines on instances; worker breach scan
Billing	plans, subscriptions, invoices
Audit	append-only audit_logs
Notifications	in-app, email, push tokens

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7. Deployment topology



Concern	Guidance
Processes	API and workers deploy separately
Health	`/health/live` , `/health/ready`
Target RPO/RTO	Documented in engineering DR notes (PITR / rollback)
Interim hosts	Railway API+worker; static web on Vercel/CDN

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8. Integration architecture

Peer	Pattern
University Platform Access	JWKS verify; `/sso/callback` handoff
Billing	Webhooks Stripe/PayMongo
OCR/AI	Provider adapters on intelligence queue
Email/Push	SMTP + FCM from notification worker

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9. Quality attributes

Attribute	Approach
Scalability	Stateless API; queue offload; optional Meilisearch

Availability	Multi-AZ DB target; worker isolation
Security	HTTPS, Shield, RBAC, rate limits, audit
Observability	Optional Prometheus `/metrics`, Sentry, structured logs

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End of Architecture Design — UDMS

Part 3 — Features & Requirements

UDMS — University Document Management System

Field	Value
Document ID	FR-UDMS-001
Product	UDMS
Companions	UDMS-SRS.md · UDMS-LLD.md
Date	August 2026

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1. Feature summary

Area	Features
Documents	Register, version, classify, attach files, tracking numbers, retention, legal hold
Workflows	Visual designer; sequential / parallel / conditional steps; publish & archive
Approvals	Queue, approve / return / reject / forward, routing timeline
SLA	Per-step deadlines, breach detection, supervisor alerts, ops metrics
Records desk	Request/release, digitization + OCR QA, receiving, archives, disposal certificates
ISO / ESS / Mobile	ISO 9001 controlled docs; employee personal records portal; mobile offline approvals
Search	Command palette, full-text + OCR, filters, optional image search
Notifications	Email, in-app, push; realtime when Socket.IO enabled
AI	Permission-aware Help Desk copilot; OCR / classification / summary pipeline
Public track	Unauthenticated status page + QR for Public documents
Admin / billing	Users, roles, branding, domains, plans, usage limits, super-admin portal
Security	Local JWT, OAuth, MFA, University Platform Access SSO, RBAC, audit logs

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2. Feature catalog (by module)

2.1 Document management

Feature ID	Feature	Description	Priority
F-DOC-01	Tracking numbers	Auto `{PREFIX}-YYYY-NNNNNNN` per tenant	Must
F-DOC-02	Lifecycle states	draft → pending_approval → approved/rejected/returned → completed → archived	Must
F-DOC-03	Draft privacy	Only creator sees drafts until submit	Must
F-DOC-04	Versioning	v0.x drafts; v1+ after approval	Must
F-DOC-05	Visibility	Public / Private; Public enables external track	Must
F-DOC-06	Confidentiality	Confidential / Restricted with watermarked download	Should
F-DOC-07	Attachments	Multipart + presigned upload; size/quota limits	Must
F-DOC-08	Retention	NAP-aligned schedules; due-date computation	Must
F-DOC-09	Legal hold	Blocks disposal while hold is active	Must
F-DOC-10	E-sign (PandaDoc)	Optional PandaDoc envelopes when enabled	Could
F-DOC-11	E-sign provider matrix	DocuSign / PNKI adapters documented; PandaDoc primary; PNKI/DocuSign pluggable	Should
F-DOC-12	NAP classification	Administrative, financial/property, legal, academic, personnel, student/support, governance categories with retention/disposition	Must
F-DOC-13	Lifecycle labels	Product states map to Active / Semi-active / Inactive / Archived for NAP reporting views	Must
F-DOC-14	Records inventory	Inventories by office, series, year, classification, retention; ref#, title, dates, volume, format, location, status	Must
F-DOC-15	PDF/A capture	Digitization targets PDF/A; QA checklist; veraPDF certification is staging ops (NFR)	Should
F-DOC-16	Confidentiality levels	Public · Internal · Confidential · Highly Confidential / Restricted with RBAC + watermark	Must

2.2 Workflows & approvals

Feature ID	Feature	Description	Priority

F-WF-01	Graph workflows	Directed JSON workflows with typed steps	Must
F-WF-02	Designer	Create draft, publish, archive workflows	Must
F-WF-03	Auto default	Provision default workflow on first submit if missing	Must
F-WF-04	Assignees	Role, named user, or department head	Must
F-APR-01	Approvals queue	Dedicated queue UI for assignees	Must
F-APR-02	Actions	Approve, Return, Reject (comment), Forward	Must
F-APR-03	Timeline	Actor + timestamp routing history	Must

2.3 SLA

Feature ID	Feature	Description	Priority
F-SLA-01	Step SLA hours	Configurable per workflow step	Must
F-SLA-02	Breach worker	Periodic detection (~5 min)	Must
F-SLA-03	Alerts & KPIs	Notify supervisors; dashboard/report metrics	Should

2.4 Records desk

Feature ID	Feature	Description	Priority
F-REC-01	Record requests	View / download / physical checkout requests	Must
F-REC-02	Release desk	Approve/deny/release/return with append-only log	Must
F-REC-03	Digitization	Batches, capture, OCR confidence, QA checklist	Must
F-REC-04	Receiving	Multi-channel inbound log + acknowledgment	Must
F-REC-05	Archives	Transfer packages create → approve → complete	Must
F-REC-06	Disposal	Justified disposal + certificate; hold enforcement	Must
F-REC-07	Receiving metadata	Title, type, originating/receiving office, dates, subject/keywords, confidentiality, retention category	Must
F-ISO-01	ISO 9001 controlled docs	Policies, procedures, manuals, forms, templates under Clause 7.5: owner, review/approve, current-version enforce, revision history	Must

F-ESS-01	Employee personal records	View-only portal for appointment, travel orders, special orders, designations; download/print gated; access logged	Must
F-MOB-01	Mobile offline approvals	Dedicated mobile UI for approvers with local cache for low connectivity	Should

2.5 Search, notify, AI, public

Feature ID	Feature	Description	Priority
F-SRH-01	Global search	Documents, files, OCR text, users, workflows	Must
F-NOT-01	Multi-channel notify	Email / in-app / push on key events	Must
F-AI-01	Help Desk copilot	Permission-aware guidance; confirm destructive acts	Should
F-AI-02	Document intelligence	Async OCR / classify / summarize	Should
F-PUB-01	Public tracking	`/track/{number}` rate-limited status	Must

2.6 Tenancy, billing, auth, admin

Feature ID	Feature	Description	Priority
F-MT-01	Multi-tenant isolation	Shared DB + `tenant_id` scoping	Must
F-MT-02	Self-service signup	Tenant + owner + onboarding wizard	Should
F-BIL-01	Subscriptions	Plans, quotas, Stripe/PayMongo checkout	Should
F-AUTH-01	Local + OAuth + MFA	Email/password, Google/Microsoft, TOTP	Must
F-AUTH-02	UPA SSO	Dual/platform mode via JWKS	Must
F-ADM-01	Team & roles	Users, custom roles, permission matrix	Must
F-ADM-02	Reports & dashboard	Metrics, trends, CSV/PDF ops pack (received/released, classification, archiving/disposal due, backlog, processing time)	Must
F-ADM-03	Super admin	Tenants, plans, platform analytics	Must
F-INT-01	University channels	UPA SSO + receiving via email/portal channels; internal-comms connectors via API	Should
F-AUD-01	Immutable audit trail	Non-editable logs: view, download/print, release, modify, approve — user, time, action, reference	Must

F-CMP-01	Retention compliance	Alerts for archival/disposal due; permanent inventory; authorized disposal workflow	Must

2.7 Source DOCX traceability (Document-Management-System.docx) — 100%

Every source requirement is documented below. Status = **Delivered** (in product scope), **Planned** (specified for delivery), **Excluded** (hardware/out of product), or **Service** (contractual).

Source requirement	Feature / ID	Status
Lifecycle receive→register→digitize→classify→inventory→release→retain→archive	F-DOC-, *F-REC-*	Delivered
RA 10173 / NAP GC 5 / ISO 9001 7.5 / COA-CHED	F-DOC-08/12, F-ISO-01, NFR-CMP	Delivered
Web multi-office/campus, modular, API	F-MT-01, Architecture	Delivered
≥500 concurrent / search ≤3s	NFR-PERF-001	Delivered (target)
Receiving + registration metadata + unique ref	F-REC-04, F-REC-07, F-DOC-01	Delivered
Routing / configurable approvals	F-WF-, *F-APR-*	Delivered
Request/release + release log	F-REC-01, F-REC-02	Delivered
Digitization + OCR + QA	F-REC-03, F-AI-02, F-DOC-15	Delivered
Archival transfer + disposal	F-REC-05, F-REC-06	Delivered
Integrated e-signatures (DocuSign/PNKI)	F-DOC-10, F-DOC-11	Delivered
Mobile app offline approvals	F-MOB-01	Planned
NAP classification categories	F-DOC-12	Delivered
Records inventory by office/series/year	F-DOC-14	Delivered
Lifecycle Active/Semi-active/Inactive/Archived	F-DOC-13	Delivered
Operational + analytical dashboards PDF/CSV	F-ADM-02	Delivered
ISO 9001 controlled document module	F-ISO-01	Delivered
Personal employee records portal	F-ESS-01	Planned

Confidentiality Public→Restricted + RBAC	F-DOC-16, F-DOC-06	Delivered
Immutable audit trail	F-AUD-01	Delivered
Retention alerts + disposal authorization	F-CMP-01, F-DOC-08/09	Delivered
University portal / email / communications	F-INT-01, F-AUTH-02	Delivered
Architecture, schema, DR, manuals	Architecture + repo docs	Delivered
Training (5 sessions) / PM / UAT / migration	SVC-UDMS-* (contractual)	Service
Physical RFID / biometric hardware	—	Excluded
veraPDF-certified PDF/A gate	F-DOC-15 note	Excluded (ops certification)

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3. Requirements matrix

Req ID	Requirement	Related features	Priority
FR-DOC-001	Assign unique tracking numbers	F-DOC-01	Must
FR-DOC-002	Enforce document lifecycle states	F-DOC-02	Must
FR-DOC-003	Restrict draft visibility to creator	F-DOC-03	Must
FR-DOC-004	Version content and files	F-DOC-04	Must
FR-DOC-005	Support Public/Private visibility	F-DOC-05	Must
FR-DOC-006	Confidential/Restricted controls	F-DOC-06	Should
FR-DOC-007	Enforce upload size and storage quotas	F-DOC-07	Must
FR-DOC-008	Retention schedules + legal hold	F-DOC-08, F-DOC-09	Must
FR-DOC-009	NAP classification + inventory views	F-DOC-12, F-DOC-14	Must
FR-DOC-010	Lifecycle NAP label mapping	F-DOC-13	Must
FR-DOC-011	E-sign providers (PandaDoc + DocuSign/PNKI matrix)	F-DOC-10, F-DOC-11	Should
FR-DOC-012	Four confidentiality levels + RBAC	F-DOC-16	Must
FR-ISO-001	ISO 9001 Clause 7.5 controlled documents	F-ISO-01	Must
FR-ESS-001	Employee personal records view-only portal	F-ESS-01	Must

FR-MOB-001	Mobile offline approvals	F-MOB-01	Should
FR-AUD-001	Immutable activity audit trail	F-AUD-01	Must
FR-CMP-001	Retention compliance alerts + disposal auth	F-CMP-01	Must
FR-INT-001	Portal / email / UPA channel integration	F-INT-01, F-AUTH-02	Should
FR-WF-001	Directed workflow graphs with step types	F-WF-01	Must
FR-WF-002	Publish/archive workflows	F-WF-02	Must
FR-WF-003	Auto-provision default workflow	F-WF-03	Must
FR-APR-001	Approvals queue with required actions	F-APR-01, F-APR-02	Must
FR-APR-002	Persist routing timeline	F-APR-03	Must
FR-SLA-001	Optional SLA hours per step	F-SLA-01	Must
FR-SLA-002	Detect breaches and notify	F-SLA-02, F-SLA-03	Must
FR-REC-001..006	Records desk capabilities	F-REC-01..06	Must
FR-SRH-001	Search across indexed entities	F-SRH-01	Must
FR-NOT-001	Event notifications	F-NOT-01	Must
FR-AI-001	Permission-aware copilot	F-AI-01	Should
FR-PUB-001	Rate-limited public tracking	F-PUB-01	Must
FR-MT-001	Tenant data isolation	F-MT-01	Must
FR-ID-001	Local auth + OAuth + MFA	F-AUTH-01	Must
FR-ID-002	University Platform Access SSO	F-AUTH-02	Must
FR-ADM-001..003	Admin, reports, super-admin	F-ADM-01..03	Must

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4. Non-functional requirements (summary)

ID	Category	Requirement
NFR-SEC-001	Security	HTTPS, validation, Shield, rate limits, RBAC
NFR-SEC-002	Security	Presigned files; secrets in env; lockout
NFR-PERF-001	Performance	Search p95 < 3s @ ~500 users (target)
NFR-AVL-001	Availability	Health probes; API ≠ worker process
NFR-CMP-001	Compliance	Audit trail, retention, disposal certificates
NFR-UX-001	Usability	Modern browsers; responsive; permission menus

5. Actors & feature access

Actor	Primary features
Staff / Member	F-DOC-*, submit, search
Approver	F-APR-*, F-MOB-01
Records Officer	F-REC-*, F-ISO-01, F-CMP-01
Employee (self-service)	F-ESS-01
Admin / Owner	F-ADM-*, F-WF-*, F-BIL-* (owner), F-AUD-01
Super Admin	F-ADM-03
Public visitor	F-PUB-01

End of Features & Requirements — UDMS

Part 4 — Low-Level Design (LLD) Workflows

UDMS (University Document Management System)

Field	Value
Document ID	LLD-UDMS-001
Product	UDMS
Companion	UDMS-SRS.md
Date	August 2026

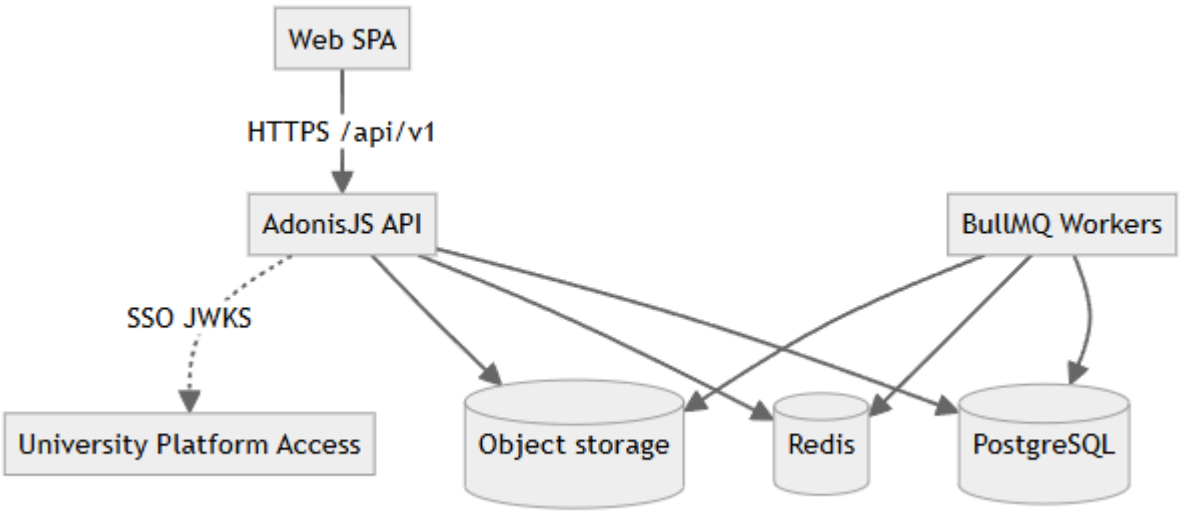
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1. Purpose

This LLD describes **runtime workflows**: actor sequences, state machines, and component interactions for UDMS. It complements the SRS with implementable process detail for developers and operators.

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2. System context (components)

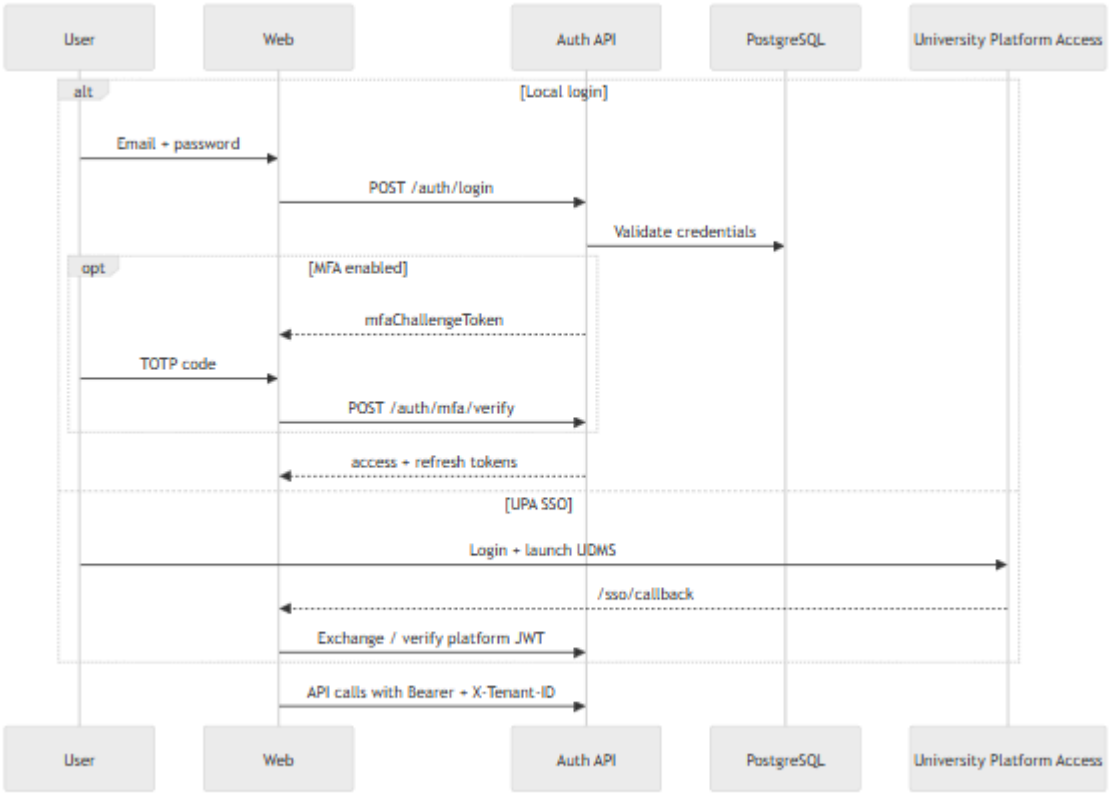


Component	Responsibility
Web SPA	UI, approvals queue, records desk, public track
API	Auth, RBAC, document/workflow services

Workers	Notifications, SLA checks, OCR/intelligence
UPA	Optional SSO issuer (dual/platform mode)

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3. Authentication workflow

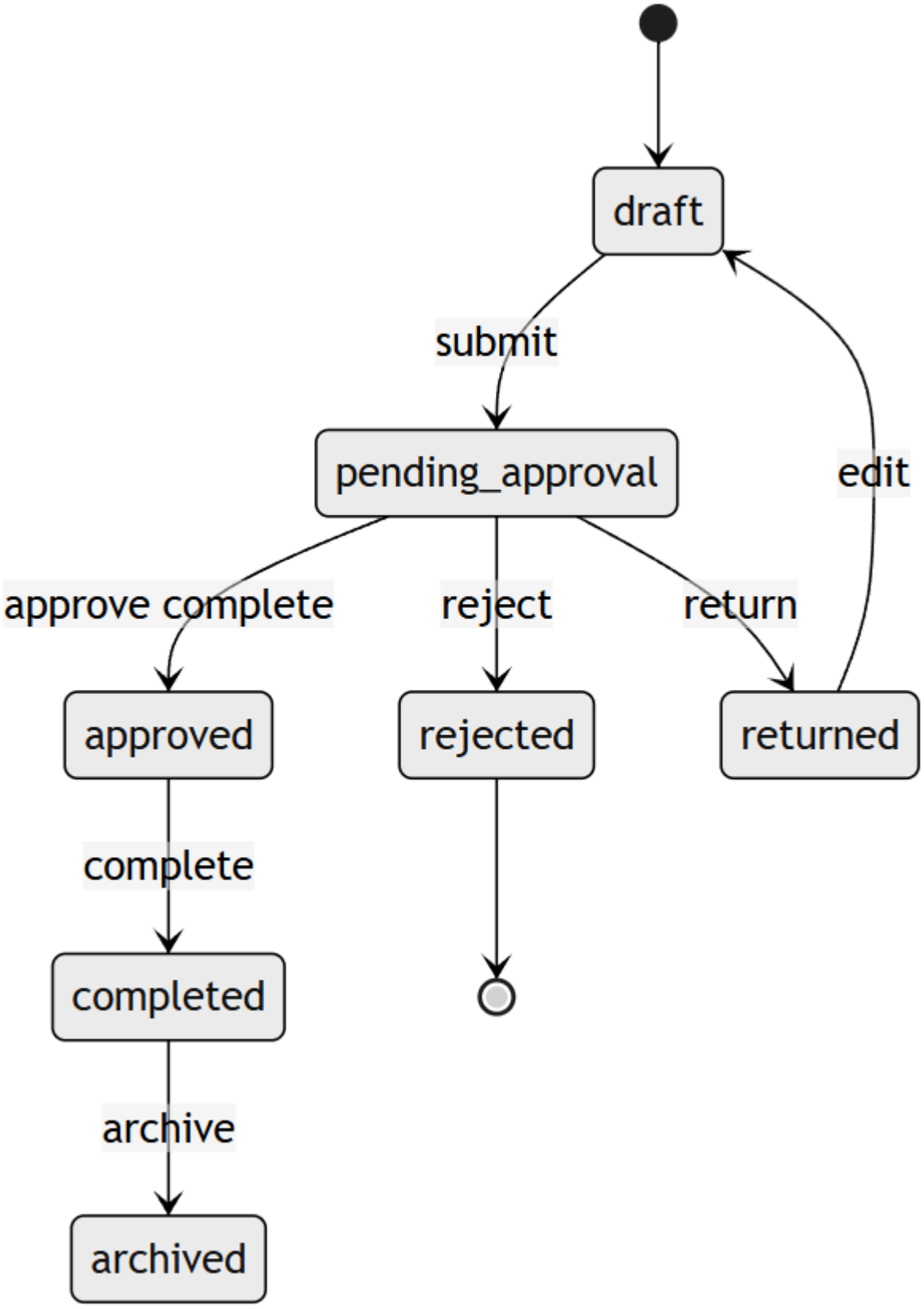


Steps

1. Resolve tenant from user membership, verified domain, or `X-Tenant-ID`.
2. Attach `TenantContext`; all queries scope by `tenant_id`.
3. Permission middleware checks RBAC (tenant owner may bypass within org).

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4. Document lifecycle (state machine)



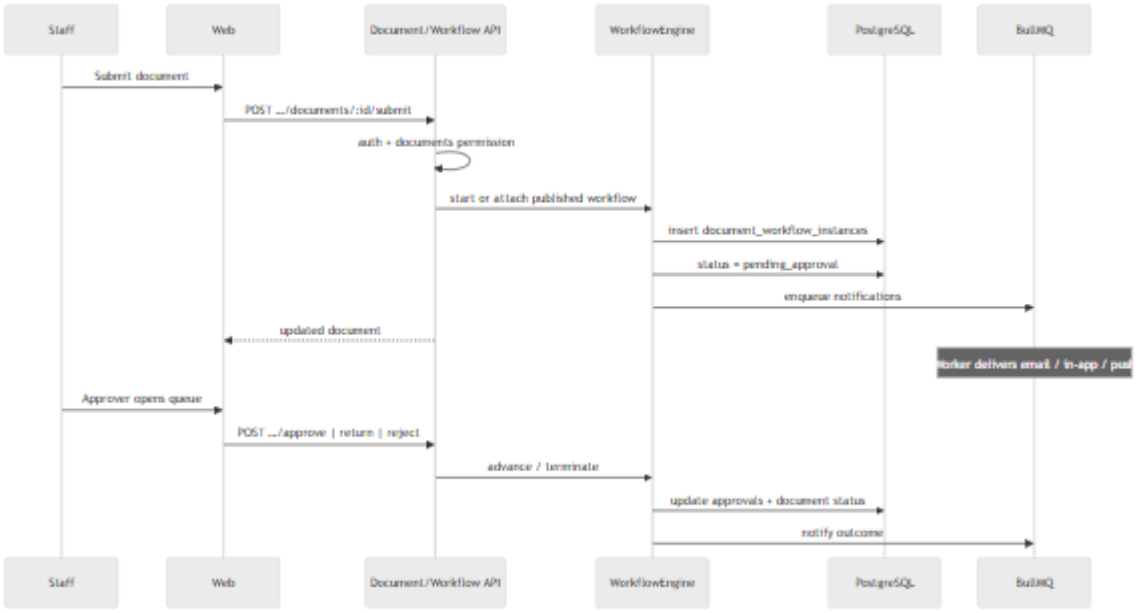
Transition	Trigger	Side effects

submit	Creator / authorized user	Tracking already assigned; start workflow instance; init SLA; notify assignees
approve	Assigned approver	Advance workflow node; may complete document
return	Approver + comment	Editable draft path; notify creator
reject	Approver + comment	Terminal; notify creator
archive	Authorized	Retention / records policies apply

Tracking number: `{PREFIX}-YYYY-NNNNNNN` (per-tenant sequence).

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5. Submit → approval pipeline



Workflow graph: JSON definition in `workflows.definition`; steps in `workflow_steps`; runtime in `document_workflow_instances.current_node_id` + `state` JSON.

Step assignment modes: role, named user, department head.

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6. SLA breach workflow

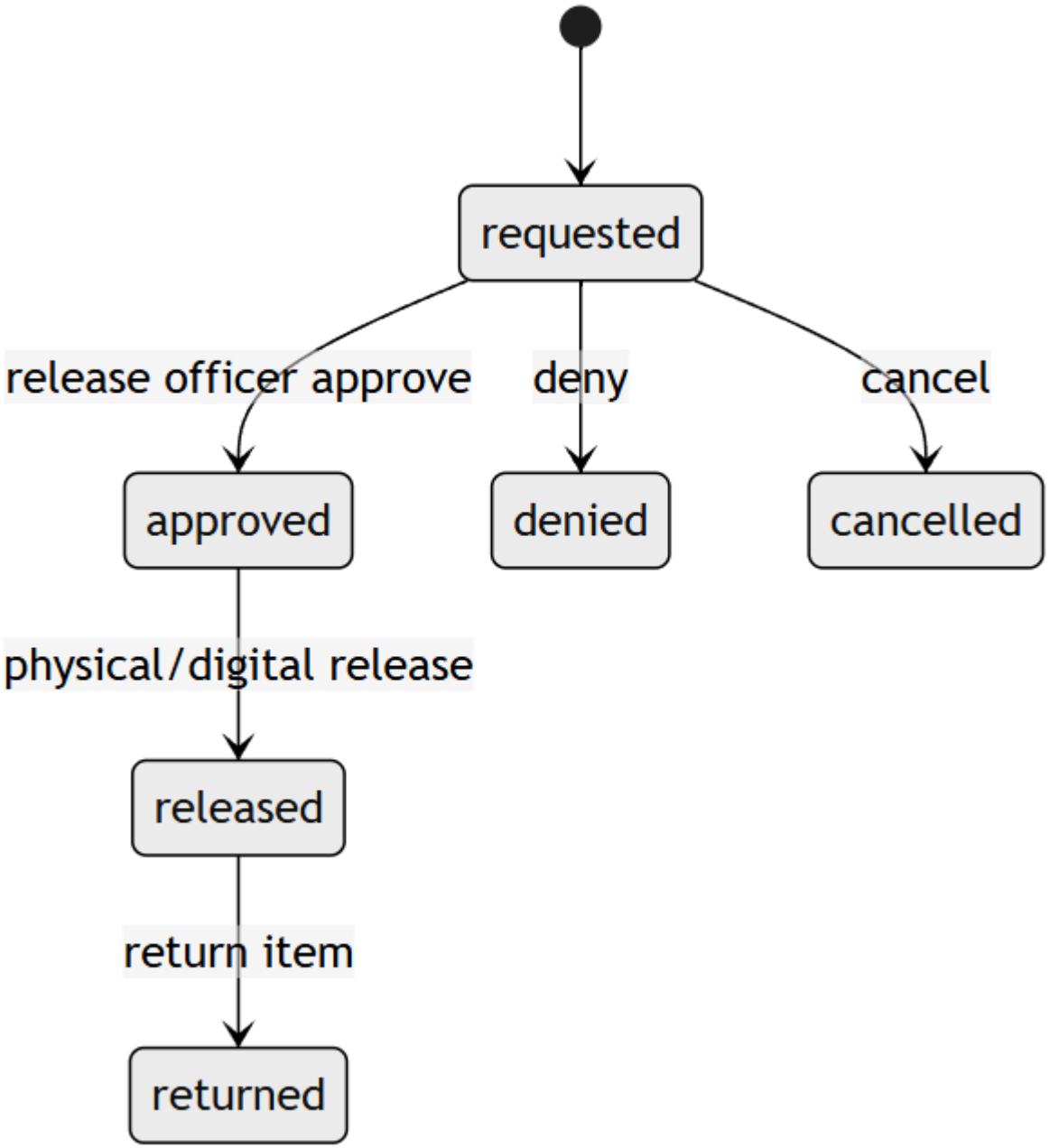


Step	Detail
1	Per-step SLA hours configured on workflow
2	SLA worker scans open instances
3	Breach writes ops signal + notification job
4	Dashboard / reports surface at-risk and breached counts

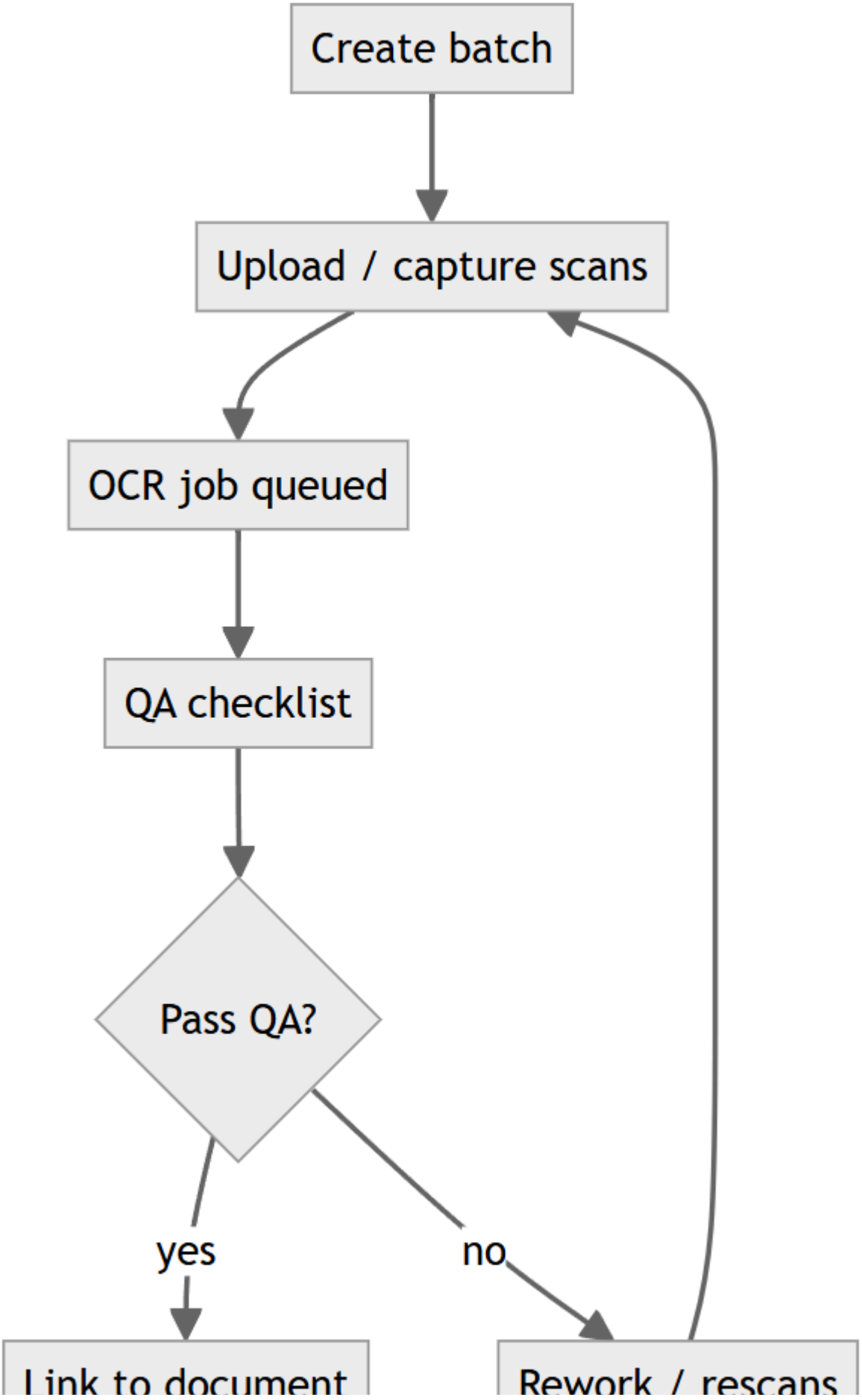
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7. Records desk workflows

7.1 Record request / release

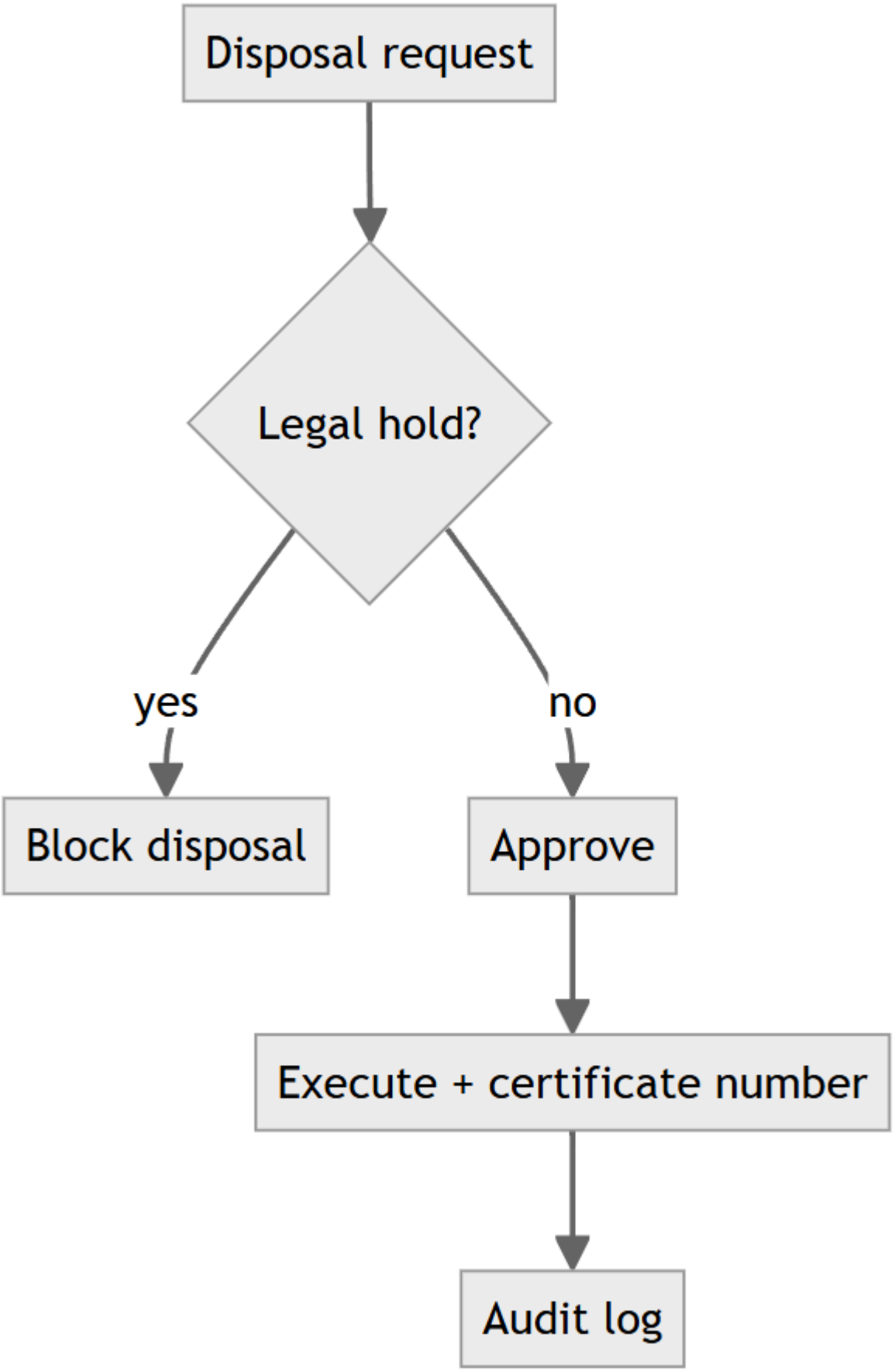


7.2 Digitization QA

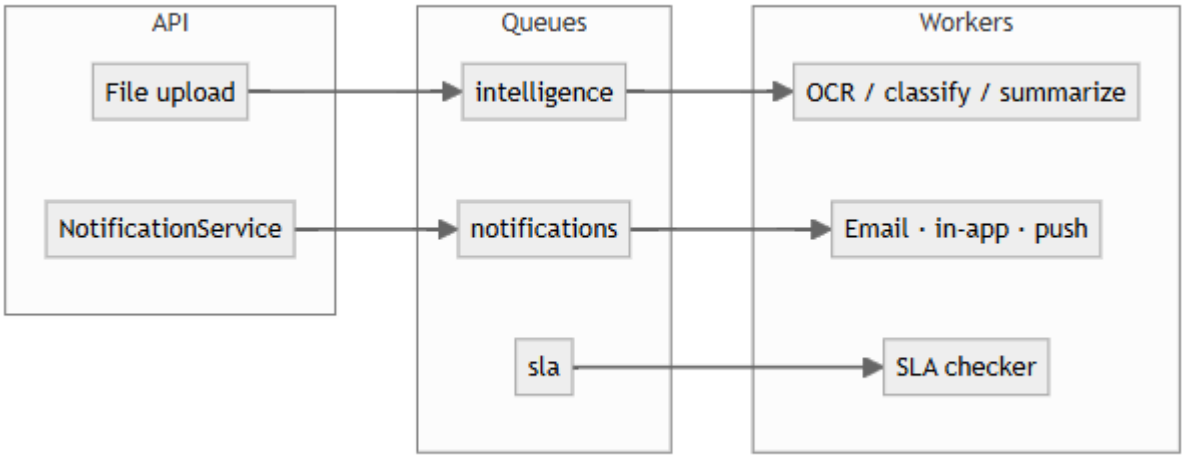

Create batch
Upload / capture scans
OCR job queued
QA checklist
Pass QA?
yes
no
Link to document
Rework / rescans



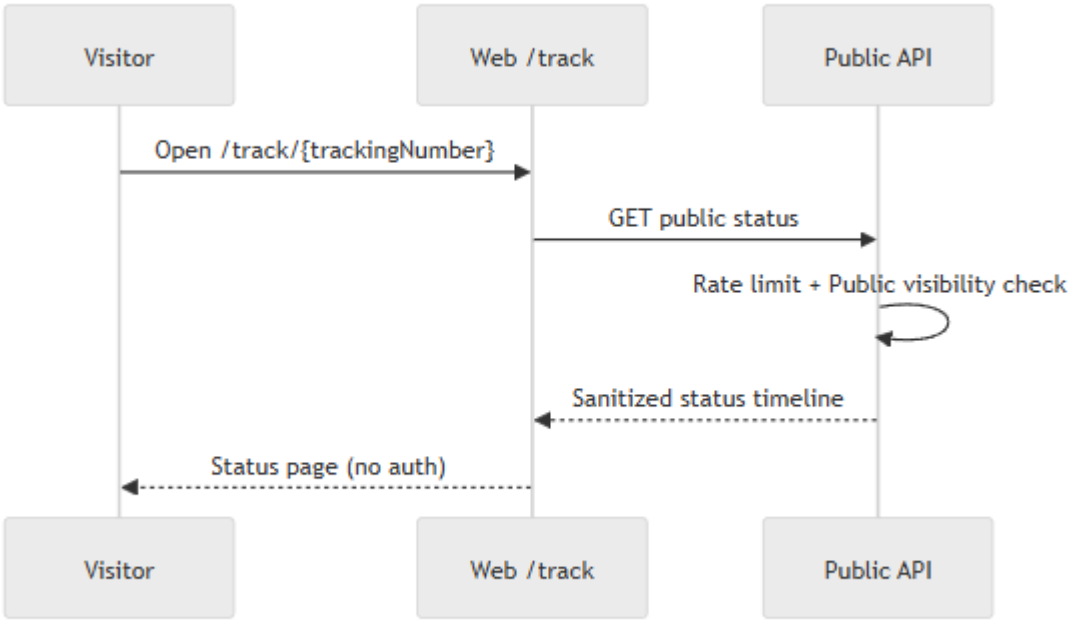
7.3 Disposal with legal hold


Disposal request
Legal hold?
yes
no
Block disposal
Approve
Execute + certificate number
Audit log

8. Async intelligence & notifications



9. Public tracking workflow



10. Error / edge cases

Case	Expected behavior
Submit non-draft	Rejected (Draft-only submit)
Approver not assignee	403
Worker down	Approvals persist; notifications/SLA/OCR lag
Plan quota exceeded	Block create user/document/storage
Restricted document	Watermark / print limits where configured

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End of LLD — UDMS
